Lab 7

Two-Dimensional Arrays in C++

Objectives

The objective of this lab is to:

- Understand what a one-dimensional array is
- Learn when to use arrays
- Study declaration and initialization of arrays
- Access and manipulate array elements
- Implement basic programs using arrays

One-dimensional array

Array is a data structure in C++, which stores a fixed size sequential collection of elements of the same type. An array is used to store a collection of data. It is often useful to think of an array as a collection of variables of the same type. All arrays consist of contiguous memory locations. The lowest address corresponds to the first element and the highest address to the last element.

A one-dimensional array is a collection of elements of the same data type stored in contiguous memory locations.

- It stores multiple values using a single variable name
- Each element is accessed using an index
- Index starts from **0**

Example:

```
int arr[5];  // Array of 5 integers
```

Declaration:

```
array_name[row][column]
```

Initialization:

It is like a variable; an array can be initialized. To initialize an array, we provide initializing values which are enclosed within curly braces in the declaration and placed following an equals sign after the array name. Here is an example of initializing an integer array.

Example:

```
int age[4]={23,34,64,75};int matrix[2][3] = {
  {1, 2, 3},
  {4, 5, 6}}
```

Example:

```
#include

using namespace std;

int main()  {

// declaration of array

int age[4];

// initialization f array elements

 age[0] = 23;

age[1] = 34;

age[2] = 64;

age[3] = 75;

return 0;

}
```

An array of 4 integers is created. Values are assigned to each variable in the array on line (8) through line (11).

Accessing Elements:

In any point of a program in which an array is visible, we can access the value of any of its elements individually as if it was a normal variable, thus being able to both read and modify its value.

Syntax:

```
arrayName[index];
```

Following the previous examples in which“age”had 4 elements and each of those elements was of type int, the name which we can use to refer to each element is the following:

Examples:

```
age[0]; // 1st element of array age[1];
        // 2nd element of array age[2];
        // 3rd element of array age[3];
        // 4th element of array Note that array index start with 0 in C++.
```

Copying Arrays

Suppose that after filling our 4 element array with values, we need to copy that array to another array of 4 integers?

```
#include 2
using namespace std;
int main() {
// declaration of array
  int age[4];
// declare first array
  int same_age[4];
// declare second array
  int i = 0;
// initialization f array elements
  age[0] = 23; age[1] = 34; age[2] = 64;
  age[3] = 75;
  for (; i < 4; i++) {
     same_age[i] = age[i]; }   //Coping one array to another
 for (i = 0; i < 4; i++) {
    cout << same_age[i] << "\t"; }
 return 0;}
```

Lab Tasks:

Task 1: Array Declaration and Initialization

- Declare an integer array called" numbers" with a size of 5.
- Initialize the array with values 10, 20, 30, 40, and 50.
- Display the values of the array elements using a loop.

```
#include <iostream>                      // Include input-output library

using namespace std;                     // Use standard namespace

int main() {                             // Main function start

   int numbers[5] = {10, 20, 30, 40, 50}; // Declare and initialize array

   cout << "Array elements are:\n";   // Display message

   for(int i = 0; i < 5; i++) {          // Loop through array

      cout << numbers[i] << " ";          // Print each element

   }

   return 0;}                             // End program
```

Task 2: Accessing Array Elements

- Declare a character array called "message" to store a word of your choice.
- Use a loop to access each character of the array and display it on a separate line.

```
#include <iostream>                      // Include library

using namespace std;                     // Use namespace

int main() {                             // Main function

   char message[] = "HELLO";             // Declare & initialize char array

   cout << "Characters are:\n";          // Display message

   for(int i = 0; message[i] != '\0'; i++) { // Loop until null character

      cout << message[i] << endl;}       // Print each cha on new line

   return 0;     }                       // End program
```

Task 3: Copying Arrays

- Declare two integer arrays: ”source” and ”destination”, each with a size of 5.
- Initialize the ”source” array with some values.
- Use a loop to copy the elements of the ”source” array into the ”destination” array.
- Display the values of both arrays to verify the copy operation.

```
#include <iostream>                      // Include library
using namespace std;                     // Use namespace
int main() {                             // Main function
    int source[5] = {1, 2, 3, 4, 5};     // Initialize source array
    int destination[5];                  // Declare destination array

    for(int i = 0; i < 5; i++) {         // Loop for copying
        destination[i] = source[i];      // Copy each element
    }
    cout << "Source Array:\n";           // Display source array
    for(int i = 0; i < 5; i++) {         // Loop through source
        cout << source[i] << " ";        // Print source elements
    }
    cout << "\nDestination Array:\n"; // Display destination array
    for(int i = 0; i < 5; i++) {         // Loop through destination
        cout << destination[i] << " ";   // Print destination elements
    }
    return 0;                            // End program
}
```

Task 4: Arrays Sum

- Write a C++ program that prompts the user to enter 5 numbers.
- Store the numbers in an integer array.
- Calculate the sum of all the numbers in the array.
- Display the calculated sum

```
#include <iostream>                      // Include library
using namespace std;                     // Use namespace
int main() {                             // Main function
  int arr[5];                            // Declare array
  int sum = 0;                           // Variable to store sum
  cout << "Enter 5 numbers:\n";          // Prompt user
  for(int i = 0; i < 5; i++) {           // Loop for input
    cin >> arr[i];                       // Store input in array
  }
  for(int i = 0; i < 5; i++) {           // Loop for sum
   sum += arr[i];                        // Add each element to sum
  }
 cout << "Sum = " << sum;                // Display result
 return 0;                               // End program
 }
```

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